Sipeng Zheng

Co-founder, BeingBeyond | Embodied AI and Multimodal Foundation Models
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I am a Co-founder of BeingBeyond, where I lead embodied multimodal pretraining and work with Prof. Zongqing Lu toward general-purpose humanoid robots. My research focuses on learning transferable perception and action representations from human experience, spanning vision-language-action models, latent world-action modeling, tactile-aware manipulation, and controllable human motion generation. I lead the Being-H, Being-M, and Being-VL model series, connecting large-scale egocentric video learning with multimodal understanding and robot control. Previously, I was a researcher at the Beijing Academy of Artificial Intelligence (BAAI). I received my Ph.D. (2023) and B.S. (2018) in Computer Science from Renmin University of China, advised by Prof. Qin Jin.

EDUCATION

Renmin University of China 2018.09 – 2023.06
Ph.D. in Computer Science *Advisor: Prof. Qin Jin*
Renmin University of China 2014.09 – 2018.06
B.S. in Computer Science *Advisor: Prof. Qin Jin*

KEY PROJECTS

Being-H0.7 & H0.8: Latent World-Action Models *2026.04 – 2026.07*
Lead the Being-H latent world-action model series, progressing from future-aware reasoning to tactile-aware robot manipulation. Being-H0.7 learns from 200,000 hours of egocentric human video and 15,000 hours of robot demonstrations, aligning a deployable latent prior with a future-informed posterior to enable direct action generation without pixel-space rollouts. Being-H0.8 extends this approach with a data foundation of 500,000+ hours of raw human video, TactoHand contact supervision, a shared TopoHand state–action space, and a slow–fast action expert for tactile feedback. **Being-H0.7: NeurIPS 2026 (Project Lead; Co-first Author).** Being-H0.8: technical report; paper forthcoming.

Being-H0 & H0.5: VLA (Project Lead; Co-first Author) *2026.01*
Built the Being-H VLA pretraining line from large-scale human videos to cross-embodiment robot control. Being-H0 learns dexterous manipulation via explicit hand-motion modeling and physical instruction tuning, while Being-H0.5 scales this direction with a unified action language and a Mixture-of-Transformers architecture. Transfers manipulation priors across heterogeneous embodiments and control spaces. **Being-H0: ICML 2026; Being-H0.5: CoRL 2026 Oral (top 1%).**

Being-M0 & M0.5: Large Motion Models (Project Lead; Co-first Author) *2025.08*
Developed real-time controllable vision-language-motion generation models trained on the HuMo100M data mixture. Being-M0.5 supports random instructions, initial-pose conditioning, long-term and unseen motion generation, and part-aware control, backed by over 5M self-collected motions and 100M multi-task instructional instances. Part-aware residual quantization enables fine-grained body-part control. **Being-M0: ICML 2025; Being-M0.5: ICCV 2025.**

Being-VL-0.5: Unified Vision-Language Model *2025.07*
Built a unified multimodal understanding framework by applying Byte-Pair Encoding-style tokenization directly to visual tokens. The method merges frequent and spatially consistent VQ token pairs into structured visual units, then trains with a curriculum that progresses from visual-language alignment to perception, reasoning, and instruction following. **Being-VL-0: ICLR 2025; Being-VL-0.5: ICCV 2025 Highlight (top 2.3%).**

Being-0: Agentic Embodied Framework *2025.02*
Integrated a foundation model with a modular skill library: high-level instruction understanding, task planning, and reasoning are handled by the foundation model, while stable locomotion and dexterous manipulation are delegated to low-level skills.

WORK EXPERIENCE

BeingBeyond 2025.05 – Present
Co-founder, Head of Embodied Multimodal Pretraining *Beijing, China*

- Lead embodied multimodal pretraining across Being-H, Being-M, and Being-VL, covering VLA and latent world-action models, tactile-aware manipulation, human motion generation, and multimodal understanding.
- Scale human-centric pretraining through data curation, hand-motion reconstruction, and cross-embodiment alignment; the latest Being-H0.8 data foundation spans 500,000+ hours of raw human video.
- Translate research into robot policies evaluated across simulation and physical platforms, including Unitree G1, PND Adam-U, and Franka FR3.

Beijing Academy of Artificial Intelligence 2023.07 – 2025.05
Researcher *Beijing, China*

- Conducted research on multimodal agents, vision-language-audio pretraining, and embodied open-world control.
- Published work on embodied agents and multimodal representation learning at ICLR, NAACL, ECCV, and related venues.

Beijing Academy of Artificial Intelligence 2022.12 – 2023.06
Research Intern *Beijing, China*

- Worked on multilingual language-vision-audio pretraining and large multimodal model research.

Microsoft Research Asia 2022.04 – 2022.10
Research Intern *Beijing, China*

- Worked on temporal sentence grounding for long-form videos.

Beijing Academy of Artificial Intelligence 2021.11 – 2022.04
Research Intern *Beijing, China*

AWARDS

Ranked 1st, GemBench Challenge at CVPR 2025 GRAIL Workshop.	2025
Ranked 3rd, Facebook CVPR 2022 Ego4D Natural Language Query Challenge.	2022
Ranked 3rd, NIST TRECVID 2021 Ad-hoc Video Search Challenge.	2021
Ranked 2nd, CVPR 2021 HOMAGE Scene-Graph Generation Challenge.	2021
Ranked 2nd, ACM MM 2020 Video Relationship Understanding Grand Challenge.	2020
Ranked 2nd, ACM MM 2019 Video Relationship Understanding Grand Challenge.	2019
National Scholarship for Ph.D. Students, Top 5%.	2022
Best Method Prize, ACM MM 2019 Grand Challenge.	2019
First-Class Scholarship for Ph.D. Students, Top 10%.	2018–2021
First Prize, National University Mathematical Modeling Competition (Beijing Area).	2015

PUBLICATIONS

† denotes equal contribution.

1. BeingBeyond Team. Being-H0.8: A Latent Tactile World-Action Model at Scale. *BeingBeyond Technical Report (paper forthcoming)*, 2026
2. Ziheng Xi, Weibo Li, Jiayi Yu, **Sipeng Zheng**, Wanpeng Zhang, Haiweng Xu, Hao Luo, Chuan Mao, and Zongqing Lu. TactoHand: Learning Geometry-Based Dense Hand-Surface Interaction Fields from Video. *arXiv*, 2026
3. Chi Zhang, Penglin Cai, Ziheng Xi, Haoqi Yuan, Hao Luo, Wanpeng Zhang, **Sipeng Zheng**, Chaoyi Xu, and Zongqing Lu. Human-Centric Transferable Tactile Pre-Training for Dexterous Robotic Manipulation. *arXiv*, 2026
4. Ye Wang†, **Sipeng Zheng**†, Hao Luo†, Wanpeng Zhang†, Haoqi Yuan, Chaoyi Xu, Haiweng Xu, Yicheng Feng, Mingyang Yu, Zhiyu Kang, Zongqing Lu, and Qin Jin. Rethinking Visual-Language-Action Model Scaling: Alignment, Mixture, and Regularization. *arXiv*, 2026
5. BeingBeyond Team. Being-H0.7: A Latent World-Action Model from Egocentric Videos. *NeurIPS*, 2026
6. Wanpeng Zhang, Hao Luo, **Sipeng Zheng**, Yicheng Feng, Haiweng Xu, Ziheng Xi, Chaoyi Xu, Haoqi Yuan, and Zongqing Lu. Conservative Offline Robot Policy Learning via Posterior-Transition Reweighting. *arXiv*, 2026
7. BeingBeyond Team. Being-H0.5: Scaling Human-Centric Robot Learning for Cross-Embodiment Generalization. *CoRL (Oral, top 1%)*, 2026
8. Boyuan Li, **Sipeng Zheng**, Bin Cao, Ruihua Song, and Zongqing Lu. Robust Motion Generation using Part-level Reliable Data from Videos. *ACM MM*, 2026
9. Haiweng Xu, **Sipeng Zheng**, Hao Luo, Wanpeng Zhang, Ziheng Xi, and Zongqing Lu. Unmasking the Illusion of Embodied Reasoning in Vision-Language-Action Models. *ECCV*, 2026
10. Wanpeng Zhang, Ye Wang, Hao Luo, Haoqi Yuan, Yicheng Feng, **Sipeng Zheng**, Qin Jin, and Zongqing Lu. DiG-Flow: Discrepancy-Guided Flow Matching for Robust VLA Models. *ECCV*, 2026

11. Junpeng Yue, Zepeng Wang, Yuxuan Wang, Weishuai Zeng, Jiangxing Wang, Xinrun Xu, Yu Zhang, **Sipeng Zheng**, Ziluo Ding, and Zongqing Lu. RL from Physical Feedback: Aligning Large Motion Models with Humanoid Control. *ECCV*, 2026
12. Bin Cao, **Sipeng Zheng**, Hao Luo, Boyuan Li, Jing Liu, and Zongqing Lu. OpenT2M: No-frill Motion Generation with Open-source, Large-scale, High-quality Data. *CVPR*, 2026
13. BeingBeyond Team. Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos. *ICML*, 2026
14. Yicheng Feng, Wanpeng Zhang, Ye Wang, Hao Luo, Haoqi Yuan, **Sipeng Zheng**, and Zongqing Lu. Spatial-Aware VLA Pretraining through Visual-Physical Alignment from Human Videos. *CVPR*, 2026
15. Hao Luo, Ye Wang, Wanpeng Zhang, Haoqi Yuan, Yicheng Feng, Haiweng Xu, **Sipeng Zheng**, and Zongqing Lu. Joint-Aligned Latent Action: Towards Scalable VLA Pretraining in the Wild. *CVPR*, 2026
16. Hao Luo, Zihao Yue, Wanpeng Zhang, Yicheng Feng, **Sipeng Zheng**, Deheng Ye, and Zongqing Lu. OpenMMEgo: Enhancing Egocentric Understanding for LMMs with Open Weights and Data. *NeurIPS*, 2025
17. Boshen Xu, Yuting Mei, Xinbi Liu, **Sipeng Zheng**, and Qin Jin. EgoDTM: Towards 3D-Aware Egocentric Video-Language Pretraining. *NeurIPS*, 2025
18. Jiazheng Liu, **Sipeng Zheng**, Börje F. Karlsson, and Zongqing Lu. Taking Notes Brings Focus? Towards Multi-Turn Multimodal Dialogue Learning. *EMNLP*, 2025
19. Yuting Mei[†], Ye Wang[†], **Sipeng Zheng**, and Qin Jin. QuadrupedGPT: Towards a Versatile Quadruped Agent in Open-ended Worlds. *arXiv*, 2025
20. Wanpeng Zhang, Yicheng Feng, Hao Luo, Yijiang Li, Zihao Yue, **Sipeng Zheng**, and Zongqing Lu. Unified Multimodal Understanding via Byte-Pair Visual Encoding. *ICCV*, 2025 (Highlight)
21. Bin Cao[†], **Sipeng Zheng**[†], Ye Wang, Lujie Xia, Qianshan Wei, Qin Jin, Jing Liu, and Zongqing Lu. Being-M0.5: A Real-Time Controllable Vision-Language-Motion Model. *ICCV*, 2025
22. Yicheng Feng[†], Yijiang Li[†], Wanpeng Zhang, **Sipeng Zheng**, and Zongqing Lu. VideoOrion: Tokenizing Object Dynamics in Videos. *ICCV*, 2025
23. Ye Wang[†], **Sipeng Zheng**[†], Bin Cao, Qianshan Wei, Weishuai Zeng, Qin Jin, and Zongqing Lu. Scaling Large Motion Models with Million-Level Human Motions. *ICML*, 2025
24. Wanpeng Zhang, Zilong Xie, Yicheng Feng, Yijiang Li, Xingrun Xing, **Sipeng Zheng**, and Zongqing Lu. From Pixels to Tokens: Byte-Pair Encoding on Quantized Visual Modalities. *ICLR*, 2025
25. Boshen Xu, Ziheng Wang, Yang Du, Zhinan Song, **Sipeng Zheng**, and Qin Jin. EgoNCE++: Do Egocentric Video-Language Models Really Understand Hand-Object Interactions? *ICLR*, 2025
26. Boshen Xu, **Sipeng Zheng**, and Qin Jin. SPAFormer: Sequential 3D Part Assembly with Transformers. *3DV*, 2025
27. **Sipeng Zheng**, Bohan Zhou, Yicheng Feng, Ye Wang, and Zongqing Lu. UniCode: Learning a Unified Codebook for Multimodal Large Language Models. *ECCV*, 2024
28. **Sipeng Zheng**, Jiazheng Liu, Yicheng Feng, and Zongqing Lu. Steve-Eye: Equipping LLM-based Embodied Agents with Visual Perception in Open Worlds. *ICLR*, 2024 (Spotlight 5.02%)
29. Yicheng Feng, Yuxuan Wang, Jiazheng Liu, **Sipeng Zheng**, and Zongqing Lu. LLaMA Rider: Spurring Large Language Models to Explore the Open World. *NAACL*, 2024
30. Qi Zhang, **Sipeng Zheng**, and Qin Jin. No-frills Temporal Video Grounding: Multi-Scale Neighboring Attention and Zoom-in Boundary Detection. In *AAAI*, 2023
31. **Sipeng Zheng**, Boshen Xu, and Qin Jin. Open-Category Human-Object Interaction Pre-training via Language Modeling Framework. In *CVPR*, 2023
32. Boshen Xu, **Sipeng Zheng**, and Qin Jin. POV: Prompt-Oriented View-agnostic Learning for Egocentric Hand-Object Interaction in the Multi-view World. *ACM MM*, 2023
33. Ludan Ruan, Anwen Hu, Yuqing Song, Liang Zhang, **Sipeng Zheng**, and Qin Jin. Accommodating audio modality in CLIP for multimodal processing. *AAAI*, 2023
34. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Few-shot Action Recognition with Hierarchical Matching and Contrastive Learning. In *ECCV*, 2022
35. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. VRDFormer: End-to-End Video Visual Relation Detection With Transformers. In *CVPR*, 2022 (Oral 4.14%)
36. **Sipeng Zheng**, Qi Zhang, Bei Liu, Qin Jin, and Jianlong Fu. Exploring Anchor-based Detection for Ego4D Natural Language Query. In *CVPR Ego4D workshop*, 2022
37. **Sipeng Zheng**, Bei Liu, Jianlong Fu, and Wen-Huang Cheng. Anchor-Based Detection for Natural Language Localization in Ego-Centric Videos. In *ICCE*, 2023
38. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Skeleton-Based Interactive Graph Network For Human Object Interaction Detection. In *ICME*, 2020
39. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Visual Relation Detection with Multi-Level Attention. In *ACM MM*, 2019
40. **Sipeng Zheng**, Xiangyu Chen, Shizhe Chen, and Qin Jin. Relation Understanding in Videos. In *ACM MM*, 2019

PROFESSIONAL SERVICE

- Conference reviewer for CVPR, ICCV, ECCV, ACCV, NeurIPS, AAAI, ACM MM, ICLR, and ICML.
- Journal reviewer for IJCV, TCSVT, TMM, and JATS.